hello abstract

1 Setup

$$\widehat{q}_h = a_{h,(k_h)}, \qquad k_h = \lceil (n_h+1)(1-\alpha) \rceil. \tag{1}$$

2 Envelope constructions

Each of the three envelope construction methods follows the same two stage calibration procedure: the shape-discovery set S_1 is used to determine the geometric shape of the envelope, then the size scaling set S_2 is used to determine the scaling factor $\hat{t}$. Each construction assigns a scalar score, $\tau(v,m)$, to a score vector v with observed set m. A candidate class is then included in the prediction set when $\tau(v,m) \le \hat{t}$, where $\hat{t}$ is calculated on S_2 as in Section 1. For both the collapsed and radial methods, we define $\tau(v) = \inf\{t > 0 : v \in t \cdot E\}$ using an envelope E which is star shaped with respect to the origin and give a closed form expression for it. The strip method works in the same way, except that the single scaling factor $\hat{t}$ multiplies the limits of the staircase rather than dilating the whole region, so τ is defined directly as a maximum of ratios.

We first fix a class h and drop the index h. K denotes the number of coordinates that remain after removing the coordinates that are missing for every sample of S_1. For a vector $v \in [0,\infty)^K$ and observed set $m \subseteq \{1,\dots,K\}$, we write v_k, $k \in m$, for just its observed entries. We use $\varepsilon = 10^{-12}$ to prevent division by zero, and $Q_p(\cdot)$ denotes the empirical p-quantile.

Collapsed envelope. The basic envelope construction simply collapses the K-dimensional score vector to a scalar score by taking the mean over the observed coordinates, and calibrates a single quantile on S_1:

$$\bar{v} = \frac{1}{|m|}\sum_{j\in m} v_j, \qquad \tilde{q} = Q_{1-\alpha}\big(\{\bar{v}_i\}_{i\in S_1}\big), \qquad \tau(v,m) = \frac{\bar{v}}{\tilde{q}}, \qquad E = \{v : \bar{v} \le \tilde{q}\}. \tag{2}$$

This method discards the geometric information that the K coordinates provide. Two vectors with the same mean are treated identically whatever the distribution of their coordinates is, so a single coordinate with strong evidence (a very small v_k) is averaged together with all the others. The accepted region is a halfspace in the observed coordinates. It serves as a mean score baseline for the multidimensional constructions below. After calibration on S_2, a vector is accepted when $\bar{v}/\tilde{q} \le \hat{t}$, which is the same as $\bar{v} \le \hat{t}\,\tilde{q}$. Here $\hat{t}$ is the k-th smallest value of $\bar{v}_i/\tilde{q}$ over S_2, so $\hat{t}\,\tilde{q}$ is just the k-th smallest value of $\bar{v}_i$ over S_2.

Radial envelope. The radial envelope method retains the directional information of the scores. It works by calibrating a boundary radius as a function of direction on the unit sphere, which is restricted to the positive orthant since all nonconformity scores are nonnegative. We sample M directions $\{u_r\}_{r=1}^M$ uniformly from the positive orthant by drawing independent vectors $g_r \sim |\mathcal{N}(0, I_K)|$ and normalizing them, $u_r = g_r/\|g_r\|$. For each vector $v_i \in S_1$, we compute its norm $\|v_i\|$ and unit direction $d_i = v_i/\|v_i\|$ using only the observed coordinates. For each direction u_r, we collect the n_r points of S_1 within an angular sector of half width $\delta = 30^\circ$, that is, those with $d_i \cdot u_r \ge \cos\delta$. The directional radius $\tilde{q}_r$ is then calibrated as

$$\tilde{q}_r = \begin{cases} Q_{1-\alpha}\big(\{\|v_i\| : d_i \cdot u_r \ge \cos\delta\}\big) & \text{if } n_r \ge n_{\min}, \\ w_r\, Q_{1-\alpha}\big(\{\|v_i\| : d_i \cdot u_r \ge \cos\delta\}\big) + (1-w_r)\,\tilde{q}_{\text{marg}} & \text{if } 0 < n_r < n_{\min}, \\ \tilde{q}_{\text{marg}} & \text{if } n_r = 0, \end{cases} \tag{3}$$

where $n_{\min} = \max\{10, \lfloor 0.2\,|S_1| \rfloor\}$ is the minimum number of points required, $w_r = n_r / n_{\min}$, and $\tilde{q}_{\text{marg}} = Q_{1-\alpha}(\{\|v_i\|\}_{i \in S_1})$ is the marginal fallback radius. This helps us avoid a hard cutoff between sufficient and insufficient local support. To get a smooth boundary, we compute the radius in an arbitrary direction d as a weighted average of all M calibrated radii,

$$R(d) = \sum_{r=1}^{M} \omega_r(d)\, \tilde{q}_r, \qquad \omega_r(d) = \frac{\exp(\kappa\, d \cdot u_r)}{\sum_{r'=1}^{M} \exp(\kappa\, d \cdot u_{r'})}, \tag{4}$$

where $\kappa > 0$ controls the angular smoothness (a larger κ gets closer to nearest-direction assignment). We evaluate R at the directions $a_t \in \{u_1, \ldots, u_M, e_1, \ldots, e_K\}$, where the e_j are the coordinate axes. This gives us the boundary points $b_t = R(a_t)\, a_t$. The envelope E is the union of the boxes $\{v : 0 \le v \le b_t\}$ over all t, and the scaling function has the closed form:

$$\tau(v, m) = \min_{t} \max_{j \in m} \frac{v_j}{b_{tj} + \varepsilon}, \tag{5}$$

which is $\inf\{t > 0 : v \in t \cdot E\}$ computed on the observed coordinates. We use $M = 250$ and $\kappa = 8$ by default.

Strip envelope. The strip envelope works by defining conditional bounds for each target coordinate i based on the bin occupied by another coordinate j. We partition the range of the observed S_1 values of coordinate j into NB equal width bins, and write $\beta_j(x)$ for the bin that contains x (a value below the smallest or above the largest observed S_1 value of coordinate j is put in the first or last bin, respectively). We work with the level $\alpha' = \alpha / K$, a Bonferroni correction for the K coordinates. It only affects the shape of the envelope, since the coverage is controlled by $\hat{t}$. For each bin n containing $n_{j,n}$ points of S_1 with both v_j and v_i observed, the upper limit along coordinate $i \neq j$ is calibrated as

$$L_{j,i,n} = \begin{cases} Q_{1-\alpha'}\big(\{v_i : \beta_j(v_j) = n\}\big) & \text{if } n_{j,n} \ge n_{\min}, \\ w_{j,n}\, Q_{1-\alpha'}\big(\{v_i : \beta_j(v_j) = n\}\big) + (1 - w_{j,n})\, \tilde{q}_i & \text{if } 0 < n_{j,n} < n_{\min}, \\ \tilde{q}_i & \text{if } n_{j,n} = 0, \end{cases} \tag{6}$$

where $\tilde{q}_i$ is the marginal $(1 - \alpha')$ quantile of coordinate i over S_1 and $w_{j,n} = n_{j,n} / n_{\min}$ shrinks the limit of underpopulated bins towards it, which is similar to the blending rule of the radial envelope. Then there is a monotonicity pass that makes sure that the limits are nonincreasing in the bin index,

$$L_{j,i,n} \leftarrow \max_{n' \ge n} L_{j,i,n'}. \tag{7}$$

This ensures that a vector with a smaller value (more confident) in the conditioning coordinate j never receives a tighter bound on coordinate i than a vector with a larger value in j. The envelope E is a coordinatewise monotone staircase, which is the intersection of these pairwise constraints. If a coordinate is unobserved, we ignore all the pairwise constraints involving it, so it is neither used as a target nor as a conditioning coordinate. The scaling function is

$$\tau(v, m) = \begin{cases} \max\limits_{\substack{j, i \in m \\ j \neq i}} \dfrac{v_i}{L_{j,i,\beta_j(v_j)} + \varepsilon}, & |m| \ge 2, \\ \dfrac{v_j}{\tilde{q}_j + \varepsilon}, & m = \{j\}, \end{cases} \tag{8}$$

so a vector is accepted when $v_i \le \hat{t}\, L_{j,i,\beta_j(v_j)}$ for every pair of observed coordinates. The single scalar $\hat{t}$ multiplies all the limits, while the bins stay fixed. We use $NB = 8$ and $n_{\min} = 3$ by default.

Properties and missing scores. The three scaling functions share one structural property.

Proposition 2.1. *Fix an observed set m and let v, v' be score vectors with observed set m such that $v'_j \le v_j$ for all $j \in m$. Then $\tau(v', m) \le \tau(v, m)$ for each of the three constructions. In particular, every accepted region $\{\tau \le \hat{t}\}$ is downward closed, so if a candidate class is accepted, it stays accepted when its scores improve.*

Proof. For (2) this is clear, because $\tilde{q}$ does not depend on v and the mean is nondecreasing in each coordinate. In (5) each ratio $v_j/(b_{tj} + \varepsilon)$ is nondecreasing in v_j, and a maximum and a minimum of nondecreasing functions is nondecreasing. For (8), $v'_j \le v_j$ gives $\beta_j(v'_j) \le \beta_j(v_j)$, so $L_{j,i,\beta_j(v'_j)} \ge L_{j,i,\beta_j(v_j)}$ by (7). Together with $v'_i \le v_i$, every ratio in (8) gets smaller. The case $|m| = 1$ is again a ratio that is nondecreasing in v_j. □

The proposition is not needed for the coverage statement of Section 1. For fixed S_1 (and fixed directions u_r, which are drawn independently of S_2 and of the new observation), each τ is a fixed function of (v, m), so the same rank argument applies to all three. Monotonicity is only a sanity property of the learned shapes.

For a vector with observed set m, the collapsed method averages over m, and the radial and strip methods only use the coordinates in m. If no coordinate is observed, the collapsed method rejects the candidate class, while the radial and strip methods accept it. During fitting we require every vector in S_1 and S_2 to have at least one observed coordinate, so the calibration does not cover this case and we make no coverage claim for it.